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Bootcamp BI Batch 18

“Customer Uplift Analysis Using Machine Learning Models”

Link Dataset :

https://www.kaggle.com/datasets/davinwijaya/customer-retention/data

Introduction — Business Problem

Context

Marketing campaigns often waste budget by sending promotions to customers who would convert anyway or who will not respond.

Problem

Traditional response models predict *who will convert*, but **do not measure whether a promotion actually causes the conversion**.

Objective

This project applies **Uplift Modeling** to identify **customers who are most likely to convert *because of* a promotion**, enabling **smarter and more cost-effective targeting.**

DATA PROFILING

Dataset Summary :

- Customer-level transaction and behavioral data
- Promotion types: **Discount** and **Buy One Get One (BOGO)**
- Binary outcome: **Conversion (0 = No, 1 = Yes)**
- Treatment variable indicating whether a customer received a promotion

Key Variables :

- Customer features (demographics / behavior / transaction history)
- Treatment group vs control group
- Conversion outcome

DATA PROFILING

Goal of Profiling :

- Understand customer distribution
- Check balance between treatment and control
- Prepare data for uplift modeling

DATA UNDERSTANDING

Treatment Variable :

Represents the type of promotion received, converted into binary flags :

- Discount = 1 / Not Discount = 0
- BOGO = 1 / Not BOGO = 0

Outcome Variable : Represents customer conversion, meaning whether the customer made a purchase.

Core Business Question : Which customers convert specifically because of a promotion, not just in general?

DATA UNDERSTANDING

Why Uplift Matters

Uplift modeling helps :

- Avoid targeting customers who would convert anyway
- Avoid wasting promotions on customers who will never convert
- Focus budget on **persuadable customers**

METHODOLOGY OVERVIEW

Step 1 — Data Preparation

- Encoded treatment types
- Created binary conversion outcome
- Split dataset into training and test sets

Step 2 — Uplift Modeling Approaches

Compared multiple uplift modeling methods :

- S-Learner — single model with treatment as a feature
- T-Learner — separate models for treated vs control groups
- Random Forest Uplift Model — capturing non-linear treatment effects

Step 3 — Uplift Prediction

- Predicted individual treatment effect (uplift) for each customer:
- Estimated difference between conversion probability with vs without promotion

MODEL EVALUATION OVERVIEW

To evaluate model quality, multiple complementary evaluation methods were used to assess both predictive behavior and business relevance.

Distribution analysis (Histogram & KDE plots) was applied to examine how uplift scores are spread across customers, helping identify whether models produce meaningful separation between high-uplift and low-uplift segments.

Scatter plots were used to observe the consistency and concentration of uplift predictions, ensuring models do not produce noisy or unstable outputs.

Qini Curve and Qini Score were then applied to measure how effectively each model ranks customers based on true incremental impact, directly reflecting potential marketing value.

This multi-layer evaluation ensures that the selected model is not only statistically sound, but also practically useful for real marketing decision-making.

S-LEARNER UPLIFT MODELLING CODING PROCESS AND VISUALIZATION EXPLANATION

Coding Process and Explanation

```
train_data, test_data = train_test_split(df, test_size=0.5, random_state=42)
```

```
# Variable Segment
var_treat = 'offer'
# Control Group
category_control = 'No Offer'
# Category Treatment = 0
category_treat_0 = 'Discount'
# Category Treatment = 1
category_treat_1 = 'Buy One Get One'
# Target Column
target = 'conversion'

# Columns To Drop From Features
drop_columns = ['conversion']

# Categorical Features To Encode
category_enc = ['zip_code', 'channel']
```

Several Data Categories Are Grouped As References For Data Search For The Later Modeling Process To Make It Easier To Carry Out The Modeling Coding Process Because The Code Syntaxes Are More Concise.

Training data is used to teach the model patterns, while **testing data** is reserved to evaluate how well the model performs on unseen data. This prevents the model from **memorizing** past data instead of **learning generalizable patterns**, ensuring the results reflect real-world performance.

A **50% train – 50% test split** was applied as a design choice to balance learning and evaluation. Other ratios are possible depending on project needs. A fixed **random_state (42)** was used to keep the data split consistent across runs, ensuring **reproducibility and stable results.**

Coding Process and Explanation

Problem: Train & Test Column Mismatch

```
train_encoded = pd.get_dummies(data=train_data, columns=category_enc, drop_first=True)
test_encoded = pd.get_dummies(data=test_data, columns=category_enc, drop_first=True)

train_encoded, test_encoded = train_encoded.align(test_encoded, join='left', axis=1, fill_value=0)
```

During categorical encoding, **training and testing datasets may contain different category values**. For example, training data may include the category **"web"**, while test data contains **"phone"**.

This can lead to **column mismatches**, causing **model input errors**. To ensure consistency, the **align() function** is used to **synchronize train and test feature columns**, following the training dataset structure (`join='left'`), while missing categories in test data are filled with **0** (`fill_value=0`). This guarantees **stable and error-free model inference**.

Coding Process and Explanation

When encoding categories such as **Web, Phone, and Multichannel**, creating all three dummy columns introduces a **mathematical dependency**:

```
train_encoded = pd.get_dummies(data=train_data, columns=category_enc, drop_first=True)
test_encoded = pd.get_dummies(data=test_data, columns=category_enc, drop_first=True)

train_encoded, test_encoded = train_encoded.align(test_encoded, join='left', axis=1, fill_value=0)
```

$$Web + Phone + Multichannel = 1 \quad \text{(always)}$$

This means **one column can always be inferred from the other two** — making it **redundant**. Keeping all three columns causes: Perfect multicollinearity, Unstable coefficients, Unreliable model interpretation. This issue is known as the **Dummy Variable Trap**.

Coding Process and Explanation

When encoding categories such as **Web, Phone, and Multichannel**, creating all three dummy columns introduces a **mathematical dependency**:

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train_encoded, test_encoded = train_encoded.align(test_encoded, join='left', axis=1, fill_value=0)
```

$$Web + Phone + Multichannel = 1 \quad \text{(always)}$$

drop_first=True Prevents Redundant Features

Instead of keeping all categories, **one column is removed** because it **does not add new information**. With `drop_first=True`,

if categories are: Web, Phone, Multichannel. We keep only: `channel_Phone`, `channel_Web`. No information is lost – the dropped category is represented implicitly.

Coding Process and Explanation

```
x_cols = train_encoded.drop(columns=[var_treat, target]).columns.to_list()
x_cols
```

```
['recency',
 'history',
 'used_discount',
 'used_bogo',
 'is_referral',
 'zip_code_Surburban',
 'zip_code_Urban',
 'channel_Phone',
 'channel_Web']
```

In this uplift modeling process, all columns are used as predictor features except for two main columns: offer and conversion. The offer column is not treated as a regular feature because it acts as a treatment, namely the type of promotion offered to customers. Meanwhile, the conversion column is used as a target, namely the outcome or customer response to be predicted.

By separating the roles of feature, treatment, and target, the model can truly learn from customer characteristics, not from promotional information or outcomes directly. This structure is crucial in uplift modeling because the primary goal is not simply to predict who will buy, but to measure the extent of the promotion's impact on customer decisions causally and fairly, without information leakage.

Coding Process and Explanation

```
array([0.04324264, 0.07230028])
```

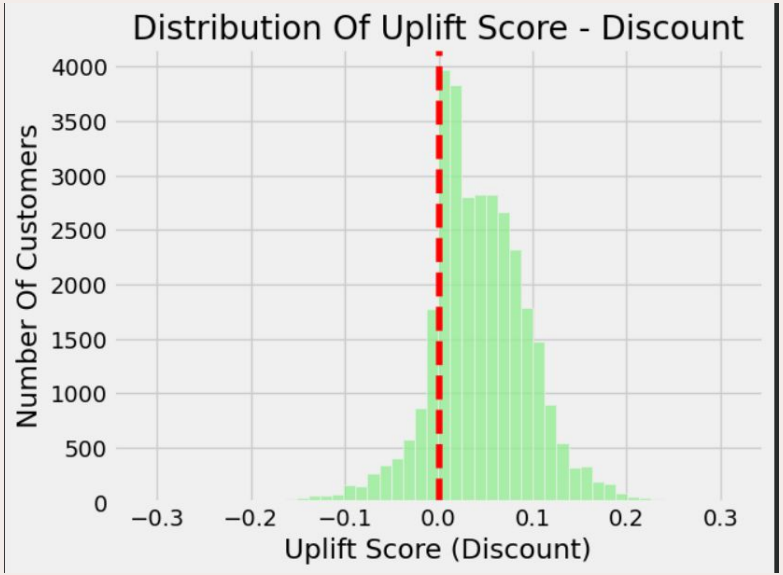
To ensure that these results were not simply due to chance in the data sample, the analysis was supplemented with bootstrap confidence intervals. The bootstrapping technique simulates repeating an experiment multiple times by resampling the data, allowing estimates not only of the average uplift value but also of the stability and confidence in that value.

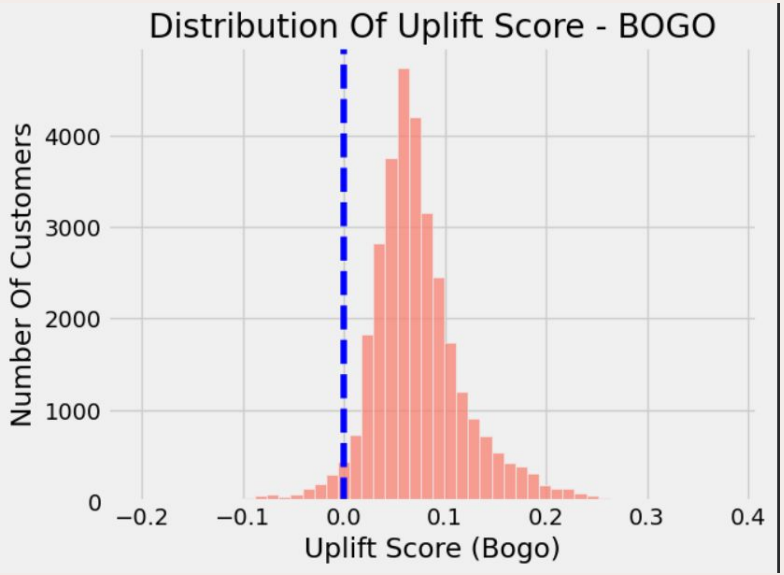
```
# ESTIMATE EFFECT
ate = S_learner.estimate_ate(X=train_encoded[x_cols].values,
                             treatment=train_encoded[var_treat],
                             y=train_encoded[target],
                             bootstrap_ci=True)

ate
```

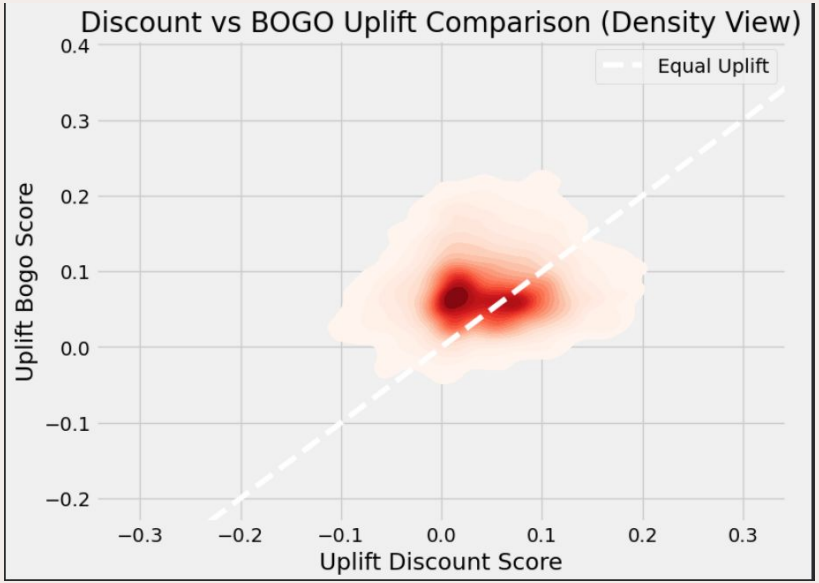
The Average Treatment Effect (ATE) results show that both types of promotions have a positive impact on customer conversion rates. Discount promotions increase the chance of conversion by an average of around 4.3%, while Buy One Get One (BOGO) promotions have a greater impact, at around 7.2%. Practically, this means that for every 100 customers receiving the promotion, Discount promotions have the potential to generate around 4 additional customers who convert, while BOGO promotions can generate around 7 additional customers compared to the no-promotion scenario.

Visualization





Visualization Explanation



Uplift density analysis shows that both Discount and Buy One Get One (BOGO) promotions generally have a positive impact on customer conversion. Most customers fall within a relatively small positive uplift area close to the equilibrium line, indicating that both promotional strategies produce similar effects for the majority of the population. No single strategy is absolutely superior across all customer segments, although BOGO shows a slight advantage in certain customer groups, which aligns with the market perception that bundled promotions often feel more attractive.

Business recommendation: Companies are advised to implement a segmentation-based promotional strategy, where BOGO is focused on customer segments with higher response potential, while Discount is used for more price-sensitive segments. This approach allows companies to maximize total uplift, improve promotional cost efficiency, and avoid suboptimal uniform promotional strategies.

RANDOM FOREST UPLIFT MODELLING CODING PROCESS AND VISUALIZATION EXPLANATION

```python
# 1. Initialization
uplift_forest = UpliftRandomForestClassifier(
    control_name=category_control,
    n_estimators=300,
    max_depth=6,
    random_state=42
)

# 2. Train (Make sure this line runs without errors)
uplift_forest.fit(
    X=train_encoded[x_cols].values,
    treatment=train_encoded[var_treat].values,
    y=train_encoded[target].values
)
```

```python
forest_predict = uplift_forest.predict(test_encoded[x_cols].values, full_output=True)
```

```python
# Build treatment binary: Discount = 1, else = 0
treatment_binary = (test_encoded[var_treat] == 'Discount').astype(int)

# Build Qini dataframe
qini_df = pd.DataFrame(data={'y':test_encoded['conversion'].astype(int),
                             'treatment': treatment_binary,
                             'uplift_discount_rf': forest_predict['delta_Discount'].values})
```

The Qini Score represents the area under the uplift curve, which mathematically measures how well the model distinguishes customers who are truly affected by the treatment from those who are not. The higher the Qini Score, the better the model's ability to capture heterogeneity in the treatment effect, meaning the model is more effective at identifying who will truly respond to the promotion.

```python
import numpy as np

qini_sorted = qini_df.sort_values(by='uplift_discount_rf', ascending=False).reset_index(drop=True)

# Outcome & treatment
outcome = qini_sorted['y'].values
treatment_flag = qini_sorted['treatment'].values

# Jumlah pelanggan treated & control secara bertahap
total_treated_so_far = np.cumsum(treatment_flag)
total_control_so_far = np.cumsum(1 - treatment_flag)

# Jumlah konversi treated & control secara bertahap
converted_treated_so_far = np.cumsum(outcome * treatment_flag)
converted_control_so_far = np.cumsum(outcome * (1 - treatment_flag))

# Estimasi uplift kumulatif
uplift_accumulated = (
    converted_treated_so_far
    - converted_control_so_far * (total_treated_so_far / np.maximum(total_control_so_far, 1))
)

# Persentase populasi
population_fraction = np.arange(1, len(outcome) + 1) / len(outcome)
```

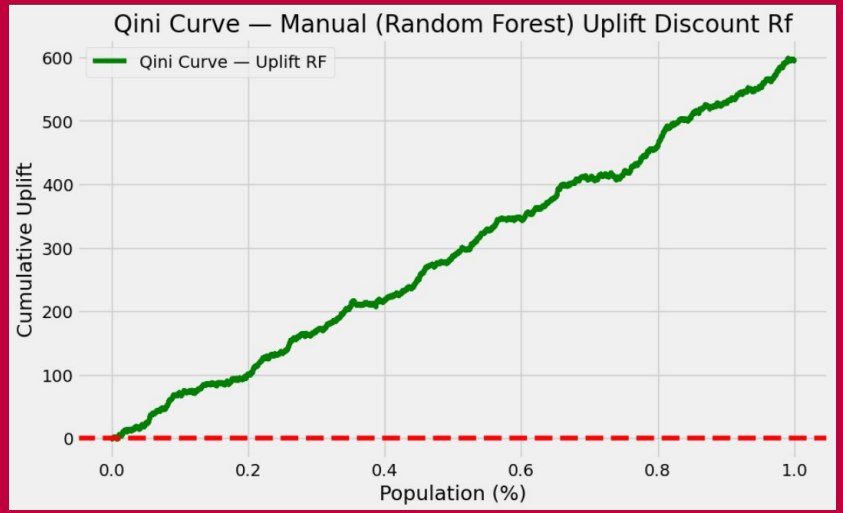
In Random Forest, max_depth controls the level of complexity of the decision tree. Too little depth results in a model that is too simple and risks underfitting, thus failing to capture variation in treatment effects. Conversely, too much depth makes the model too specific to the training data and risks overfitting, as evidenced by a Qini curve that rises sharply at the beginning but then plateaus or declines on the test data.

```python
qini_score_manual = np.trapz(uplift_accumulated, population_fraction)
print("Manual Qini Score:", qini_score_manual)
```

```
Manual Qini Score: 290.1416938714834
```

Meanwhile, n_estimators determines the number of trees in the ensemble. Too few trees can result in unstable predictions, while a larger number increases stability until saturation is reached. Therefore, the best values for max_depth and n_estimators are chosen based on the highest Qini Score on the test data, which indicates the optimal balance between model complexity and generalizability to real causal effects.

The meaning of the Qini score from the value 290.14. Technically: Qini score = area under the uplift curve. Large value = model is able to distinguish who is truly affected by the treatment. Logically: If the model is used for targeting rather than random targeting, the model produces approximately 290 additional effective conversions at this dataset scale. This means: The model has real causal value, the effect is not chance, and the model truly learns the heterogeneity of the treatment effect.



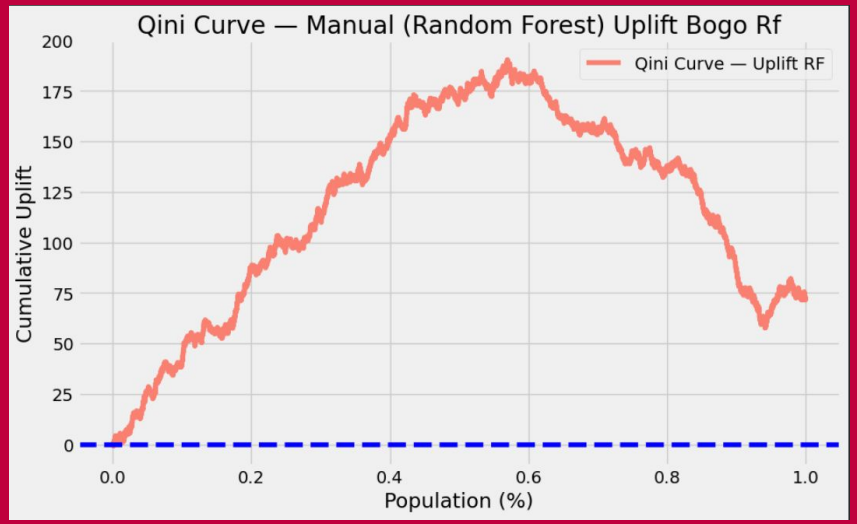
The Qini Curve shows that the Random Forest model produces stable and consistent uplift across the entire population, with the curve steadily increasing without any sharp declines. This indicates that the model is not only effective for a small segment of customers, but is capable of having a broad positive impact across a diverse customer group.

The Qini score of 290.14 indicates that the model has a strong ability to distinguish customers who are truly responsive to promotions versus random targeting, while also demonstrating that the learned uplift pattern is stable and not the result of noise or overfitting.

```python
qini_score_manual_1 = np.trapz(uplift_accumulated_1, population_fraction_1)
print("Manual Qini Score:", qini_score_manual_1)
```

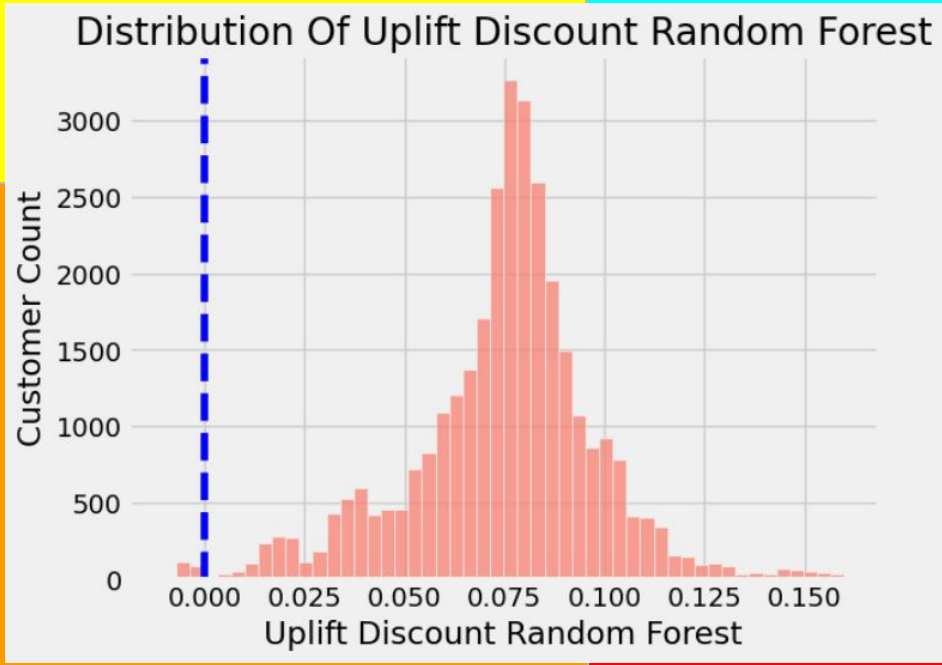
```
Manual Qini Score: 116.45908239660362
```

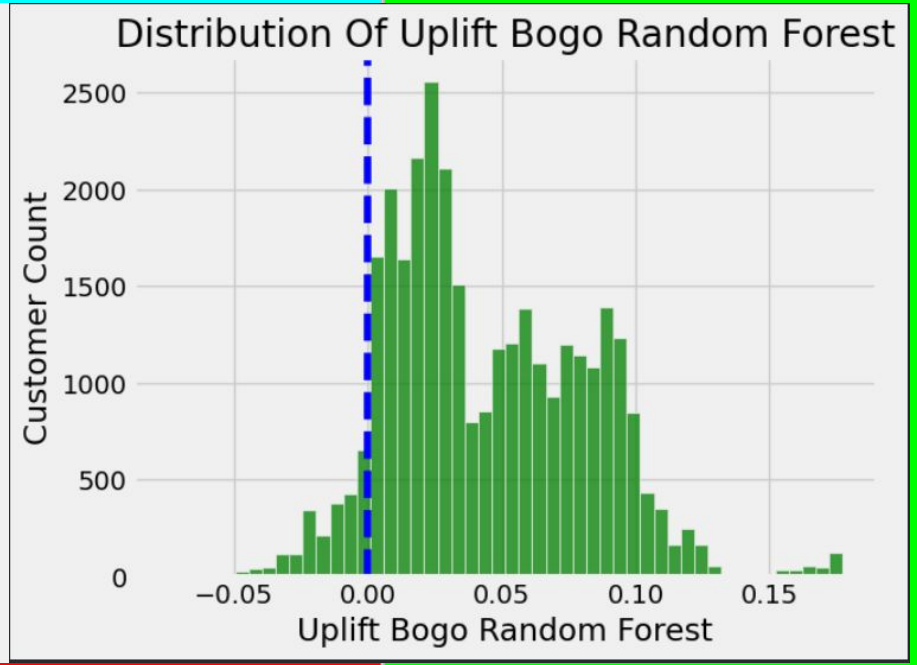
The Qini Score is 116.46. The Qini Score is the area under the uplift curve. Because the BOGO curve appears to rise and then fall, the total area is smaller. The realistic meaning is that the BOGO model generates approximately 116 incremental conversions compared to random targeting in this dataset. Compared to Discount RF (≈290), Discount is more stable and scalable. However, BOGO is sharper but more limited to specific customer segments.

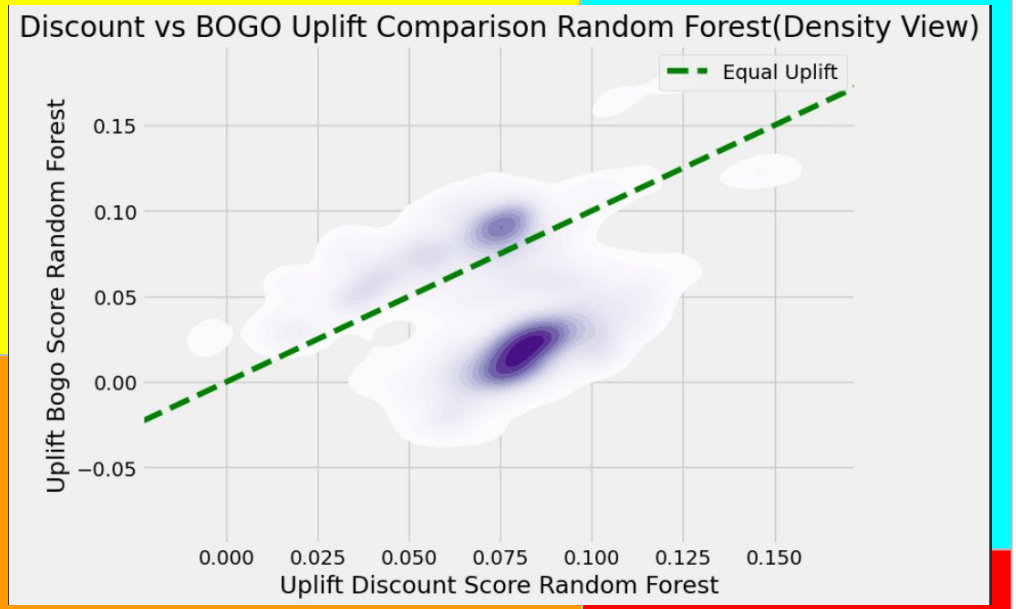


The Qini Curve for the BOGO promotion shows a sharp increase to approximately 60% of the population before experiencing a significant decline. This indicates that BOGO is highly effective for certain, highly responsive customer segments, but its effectiveness decreases when applied to broader segments.

The Qini score of 116.46 indicates that the model is able to identify customer groups that truly benefit from BOGO, but the impact is narrower and more segmented than Discount promotions. This finding implies that BOGO should be used as a targeted and selective promotional strategy, rather than aimed at the entire customer population.


Distribution Of Uplift Discount Random Forest
Customer Count
3000
2500
2000
1500
1000
500
0
0.000
0.025
0.050
0.075
0.100
0.125
0.150
Uplift Discount Random Forest


Distribution Of Uplift Bogo Random Forest
Customer Count
2500
2000
1500
1000
500
0
−0.05
0.00
0.05
0.10
0.15
Uplift Bogo Random Forest



The uplift distribution visualization shows that the majority of customers are located in areas where the uplift from Discount is higher than the uplift from BOGO. The densest concentration of data is clearly visible below the equality line (y = x), meaning that for a large portion of the population, Discount promotions have a greater impact on conversion increases than BOGO. Practically, this finding indicates that Discount strategies are more effective at driving customer response than BOGO, given the data and model used.

From a business perspective, these results have direct implications for marketing budget efficiency. Allocating more budget to Discount campaigns has the potential to generate higher incremental conversions per customer, while widespread use of BOGO campaigns risks lower uplift and reduced revenue potential. BOGO strategies should be limited to specific segments that demonstrate a positive response. Therefore, the most optimal approach is to prioritize Discount campaigns as the primary strategy, while selectively using BOGO campaigns based on customer segmentation to maximize promotional effectiveness and ROI.

T-LEARNER UPLIFT MODELLING CODING PROCESS AND VISUALIZATION EXPLANATION

```python
control_data = train_encoded[train_encoded[var_treat] == category_control]

discount_data = train_encoded[train_encoded[var_treat] == category_treat_0]

bogo_data = train_encoded[train_encoded[var_treat] == category_treat_1]

print(len(control_data), len(discount_data), len(bogo_data))
```

To analyze treatment effects accurately, the dataset was split into three groups based on promotion type: customers who received no offer (Control), customers who received a Discount offer, and customers who received a Buy-One-Get-One (BOGO) offer. This segmentation allows the model to compare customer responses across different promotional strategies and estimate the true incremental impact of each treatment. The number of records in each group was also verified to ensure balanced and reliable model evaluation.

```python
model_control = LGBMClassifier(random_state=42)
model_discount = LGBMClassifier(random_state=42)
model_bogo = LGBMClassifier(random_state=42)

model_control.fit(X=control_data[x_cols], y=control_data[target])
model_discount.fit(X=discount_data[x_cols], y=discount_data[target])
model_bogo.fit(X=bogo_data[x_cols], y=bogo_data[target])
```

Three separate LightGBM classification models were trained for each customer group: Control (no promotion), Discount, and Buy-One-Get-One (BOGO). Each model learns the conversion behavior under its specific treatment condition. This approach allows the estimation of individualized treatment effects by comparing predicted outcomes across promotional strategies, enabling more accurate uplift modeling and targeted decision-making.

```
pred_control = model_control.predict_proba(X=test_encoded[x_cols])[:,1]
pred_discount = model_discount.predict_proba(X=test_encoded[x_cols])[:, 1]
pred_bogo = model_bogo.predict_proba(X=test_encoded[x_cols])[:,1]
```

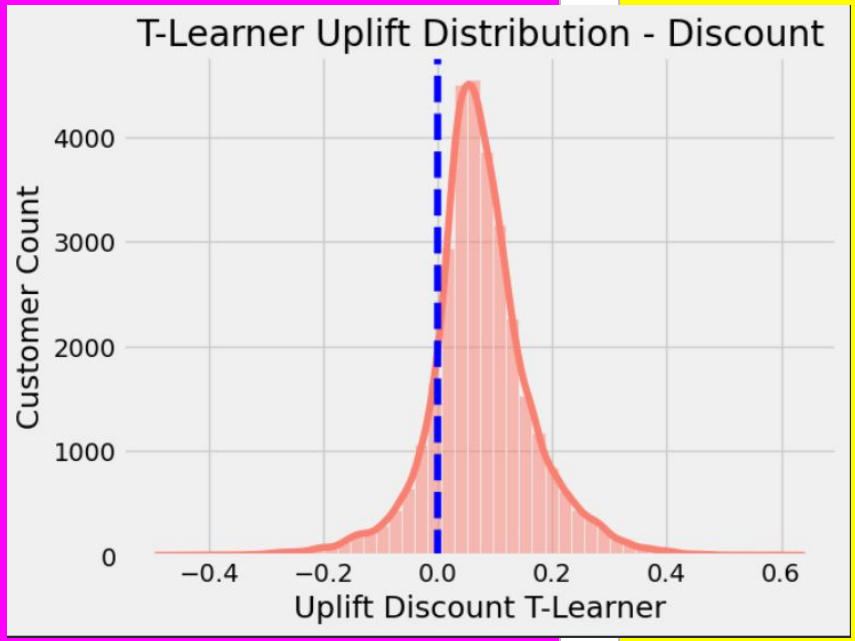
Each trained model predicts the probability of customer conversion under three different scenarios: no promotion (Control), Discount, and Buy-One-Get-One (BOGO). These probability estimates serve as the foundation for calculating uplift by comparing how the same customer is expected to respond under each promotional strategy.

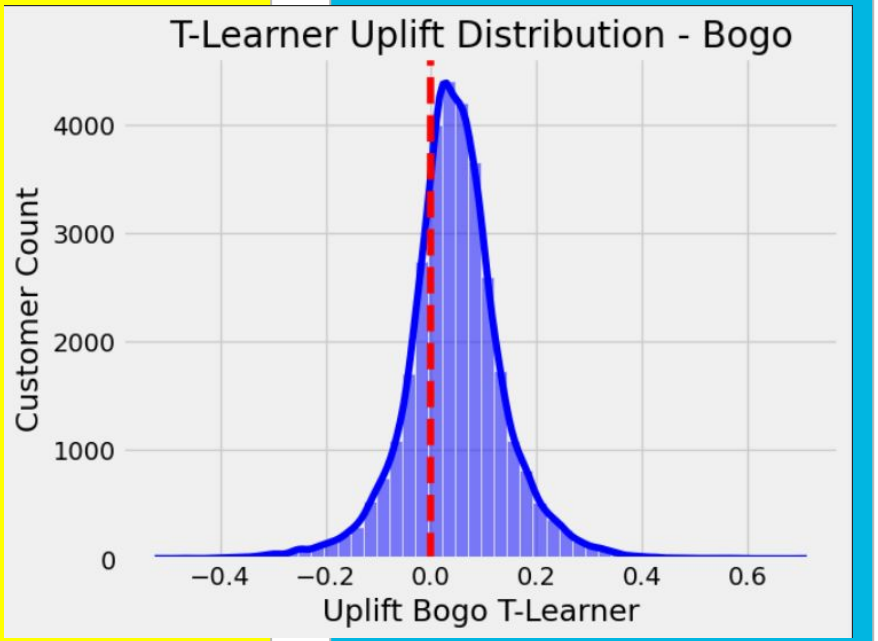
```
uplift_discount_tl = pred_discount - pred_control
uplift_bogo_tl = pred_bogo - pred_control
```

Uplift is calculated by subtracting the predicted conversion probability under the Control condition from the predicted probability under each promotional treatment. This represents the incremental impact of Discount and BOGO on customer behavior, enabling the identification of which promotion generates the greatest causal effect for each individual.

```
test_encoded['uplift_discount_tl'] = uplift_discount_tl
test_encoded['uplift_bogo_tl'] = uplift_bogo_tl
```

The calculated uplift scores for Discount and BOGO are stored in the dataset, allowing each customer to be assigned an individual-level estimated treatment effect. This enables downstream analysis, visualization, and targeted decision-making based on predicted promotional impact.


T-Learner Uplift Distribution - Discount
Customer Count
4000
3000
2000
1000
0
−0.4
−0.2
0.0
0.2
0.4
0.6
Uplift Discount T-Learner


T-Learner Uplift Distribution - Bogo
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0.0
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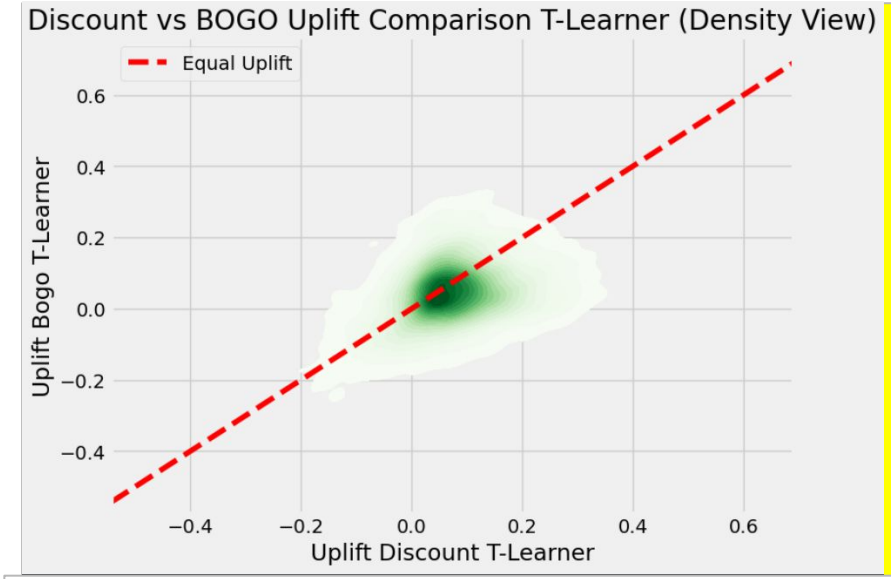
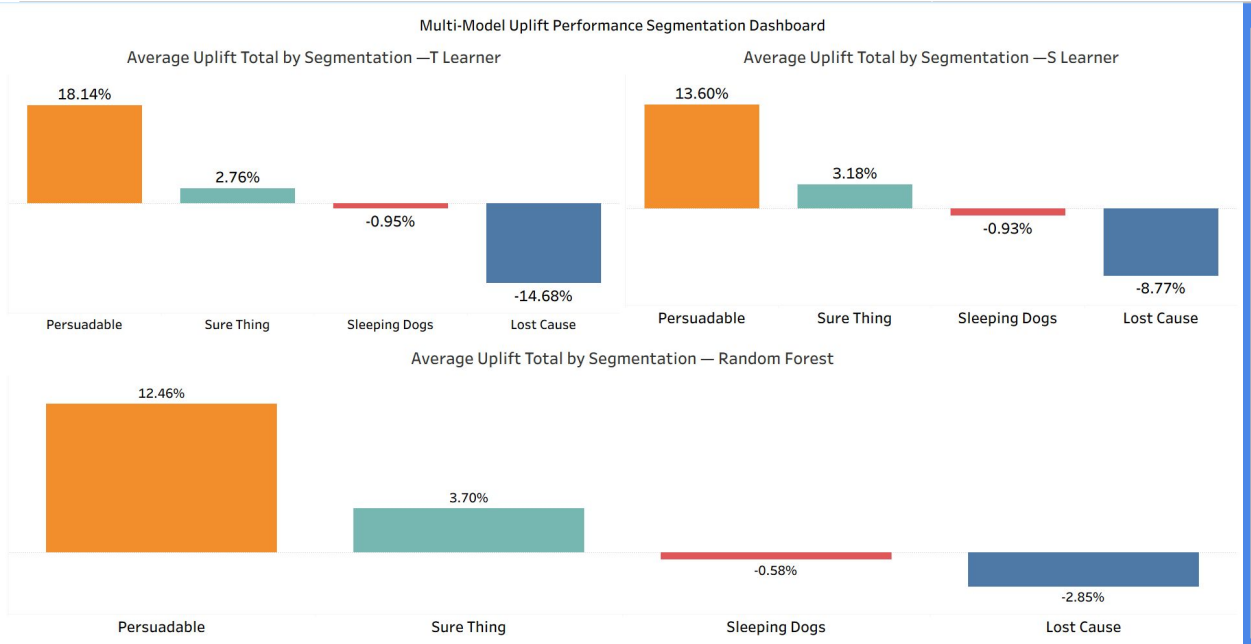
0.4
0.6
Uplift Bogo T-Learner



The uplift comparison between Discount and BOGO indicates that, on average, both promotions produce similar effects, as most customer responses cluster near the diagonal line. This suggests there is no strong overall dominance of one promotion type across the entire population. However, a small high-value segment appears in the upper-right region, showing customers who are highly responsive to both promotions—representing a persuadable and revenue-efficient target group.

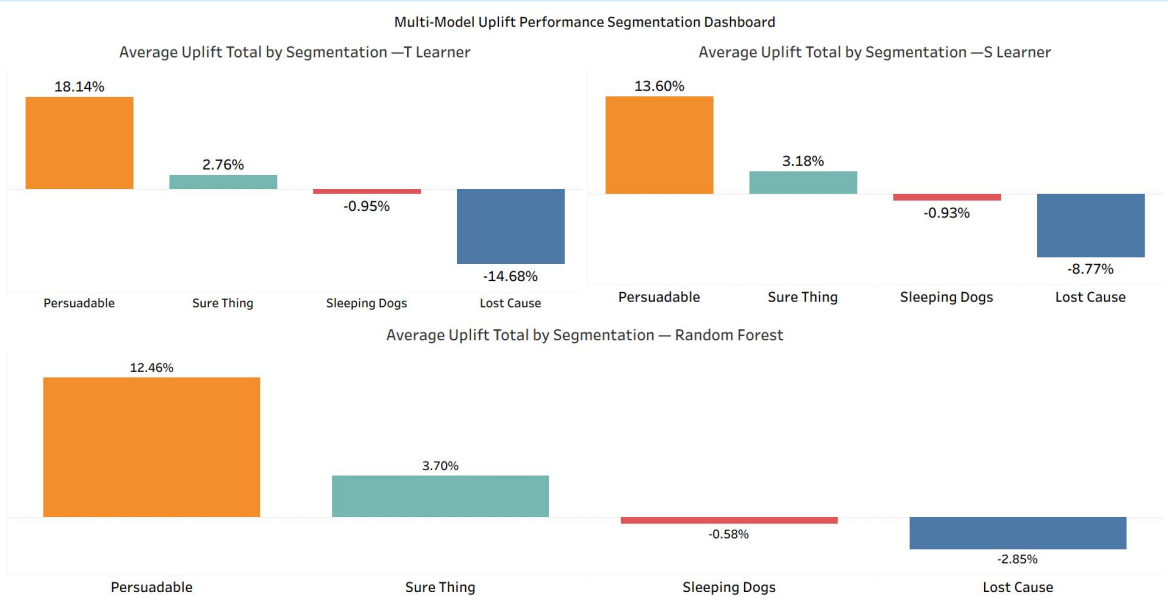
At the same time, the presence of customers in the negative uplift region implies that promotions can sometimes reduce purchase likelihood, meaning promotional offers may unintentionally cannibalize organic demand. This highlights the risk of applying mass promotions indiscriminately. From a business perspective, these results support a shift from broad, uniform promotions to an uplift-driven targeting strategy. Promotions should be selectively offered only to customers predicted to benefit, while avoiding incentives for customers likely to convert without discounts or react negatively. This approach improves marketing efficiency, reduces wasted promotional spend, and maximizes return on investment by delivering the right offer to the right customer.

VISUALIZATION, INSIGHT AND BUSINESS RECOMMENDATION BASED ON UPLIFT MODELING

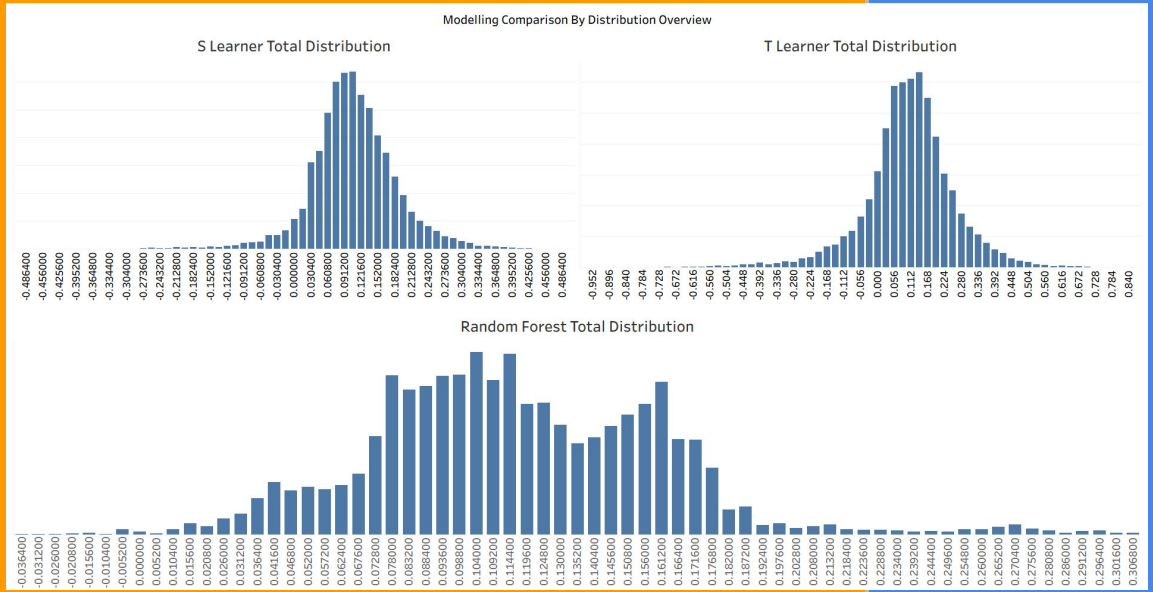


Across all three uplift models (T-Learner, S-Learner, and Random Forest), the Persuadable segment consistently shows the highest positive uplift, indicating that this group represents the most responsive and revenue-efficient customers for promotional targeting. These customers are highly influenced by incentives, meaning marketing efforts focused on this segment are likely to generate the greatest incremental conversions and return on investment.

The Sure Thing segment exhibits only a small uplift, suggesting that these customers are likely to convert even without promotions. Offering incentives to this group may result in unnecessary cost with limited incremental impact. Meanwhile, the Sleeping Dogs and Lost Cause segments show negative uplift across models, implying that promotions may reduce conversion likelihood or waste budget on customers who are unlikely to respond.

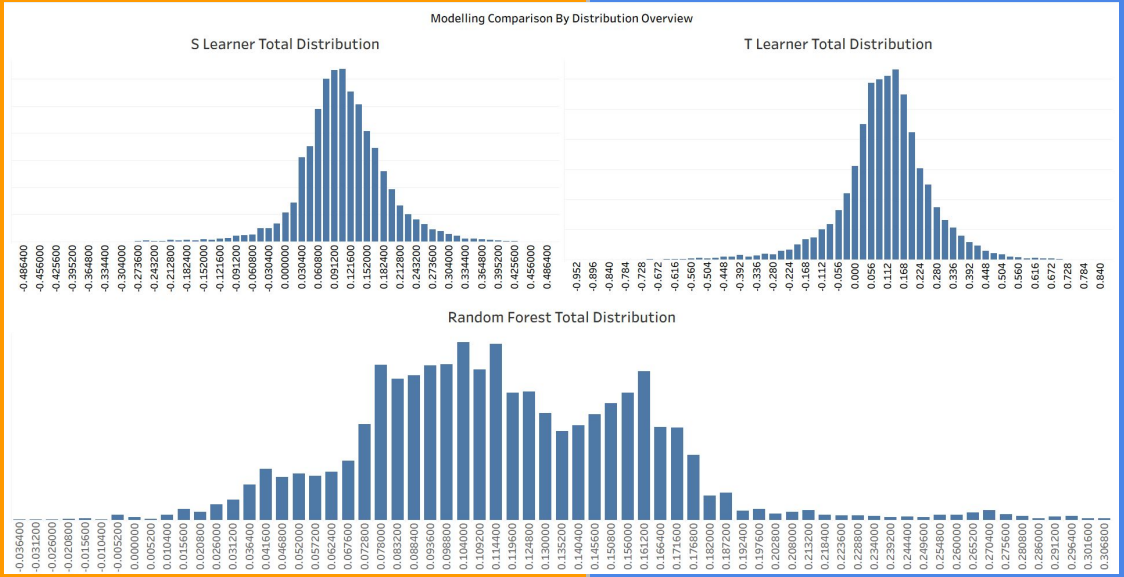


From a business perspective, these findings support a targeted promotion strategy rather than mass discounting. Marketing resources should be prioritized toward Persuadable customers, minimized for Sure Things, and avoided entirely for Sleeping Dogs and Lost Cause segments. By aligning promotional spend with uplift-based segmentation, the company can improve campaign efficiency, reduce wasted discounts, and maximize marketing ROI through more precise and data-driven decision-making.

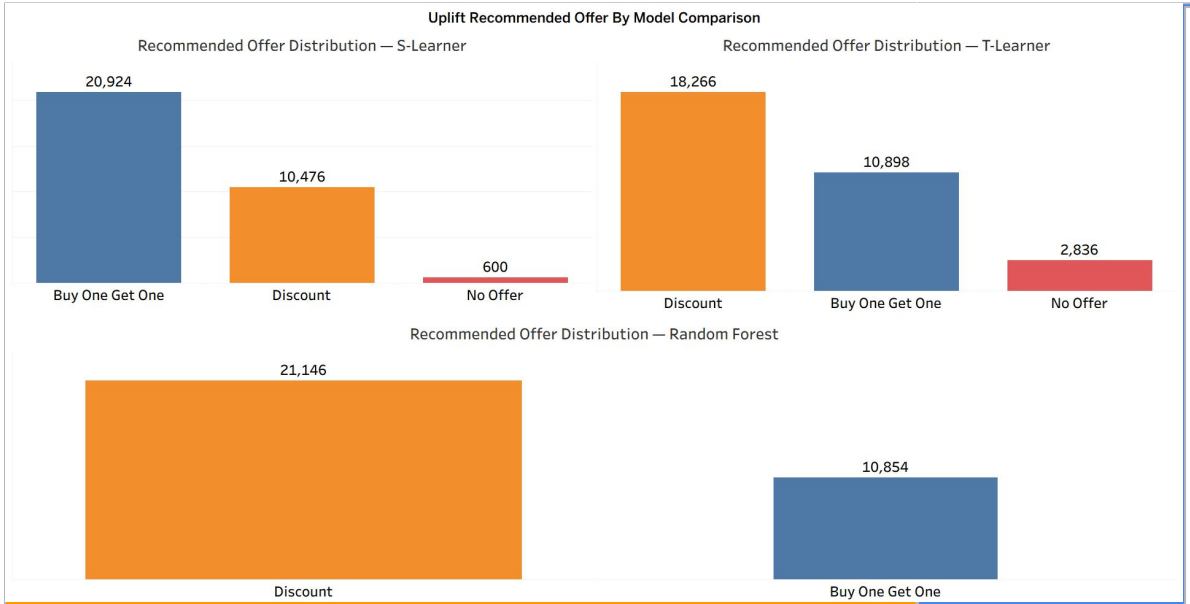


The uplift distribution across S-Learner, T-Learner, and Random Forest shows that most customers experience relatively small incremental effects from promotions, as indicated by the concentration of scores around zero. This suggests that promotional impact is not uniform across the customer base, reinforcing that mass promotion strategies are inefficient.

The T-Learner and S-Learner models produce more symmetric and tightly clustered distributions, indicating more conservative and smoother uplift estimates. In contrast, the Random Forest model generates a wider and more dispersed uplift distribution, capturing stronger heterogeneity in customer response. This implies that Random Forest is more sensitive to detecting high-potential customers as well as customers who may respond negatively to promotions.

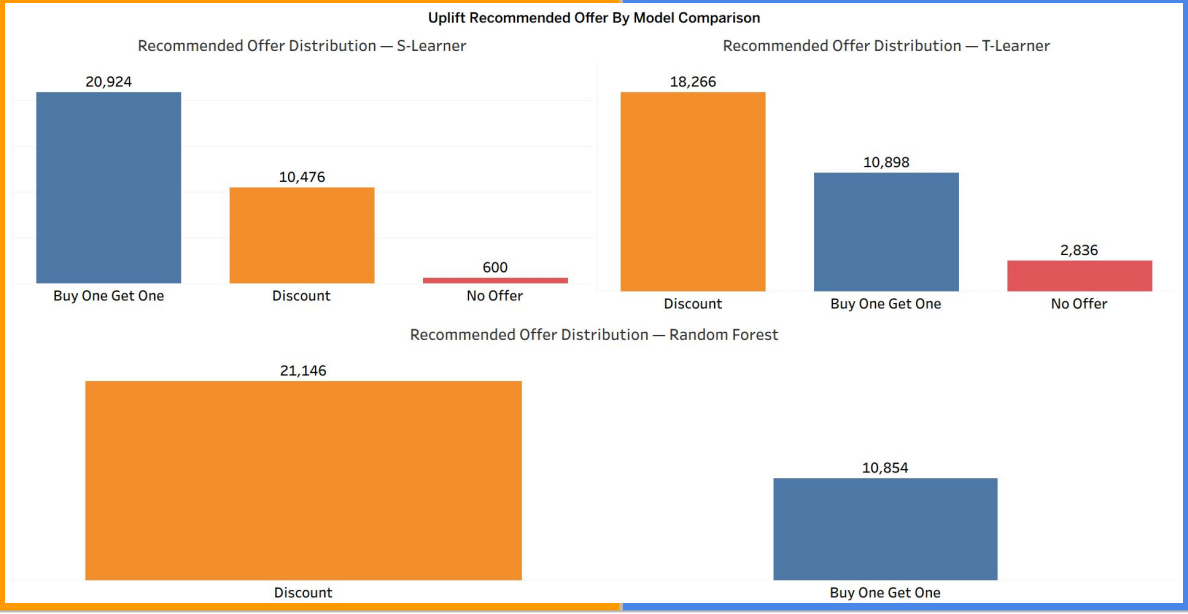


From a business standpoint, these results highlight the importance of targeted uplift-based marketing rather than blanket discounting. Marketing investments should focus on customers with clearly positive uplift scores, where promotions generate true incremental conversions, while minimizing or avoiding incentives for customers with near-zero or negative uplift to prevent wasted budget. Leveraging uplift-driven targeting can significantly improve campaign efficiency, maximize ROI, and ensure promotions are allocated only where they create real business value.



The model comparison shows meaningful differences in how promotional strategies should be allocated across customers. The S-Learner tends to recommend Buy One Get One (BOGO) more frequently, while the T-Learner and Random Forest prioritize Discounts for a larger portion of customers. This suggests that more advanced causal models detect stronger and more consistent uplift from Discount campaigns than from BOGO.

Across all models, a smaller segment of customers is recommended to receive no offer, indicating that uplift modeling can successfully identify customers who are unlikely to benefit from promotions and for whom incentives would be inefficient or unnecessary. This highlights the opportunity to reduce wasted marketing spend by withholding offers from low-impact or negative-uplift customers.



From a business perspective, the results support a shift from uniform promotions to a personalized, model-driven offer strategy. Discounts should be prioritized as the primary promotional lever for most customers, while BOGO can be reserved for specific segments where it delivers higher relative uplift. By using uplift-based recommendations, companies can optimize budget allocation, improve campaign ROI, avoid over-discounting, and ensure each customer receives the most effective offer based on predicted incremental impact.

Thank you

Link Tableau :

https://public.tableau.com/app/profile/magasa.faiz/viz/Uplift_Modelling_Visualization/UpliftRecommendedOfferByModelComparison?publish=yes

Link Google Colab :

https://drive.google.com/file/d/1dJoXrr_tc74fcksKoxfGnF6XSFiaunXF/view?usp=sharing